Borrego Springs Alluvial Fan Active and Inactive Area Mapping, County of San Diego, California

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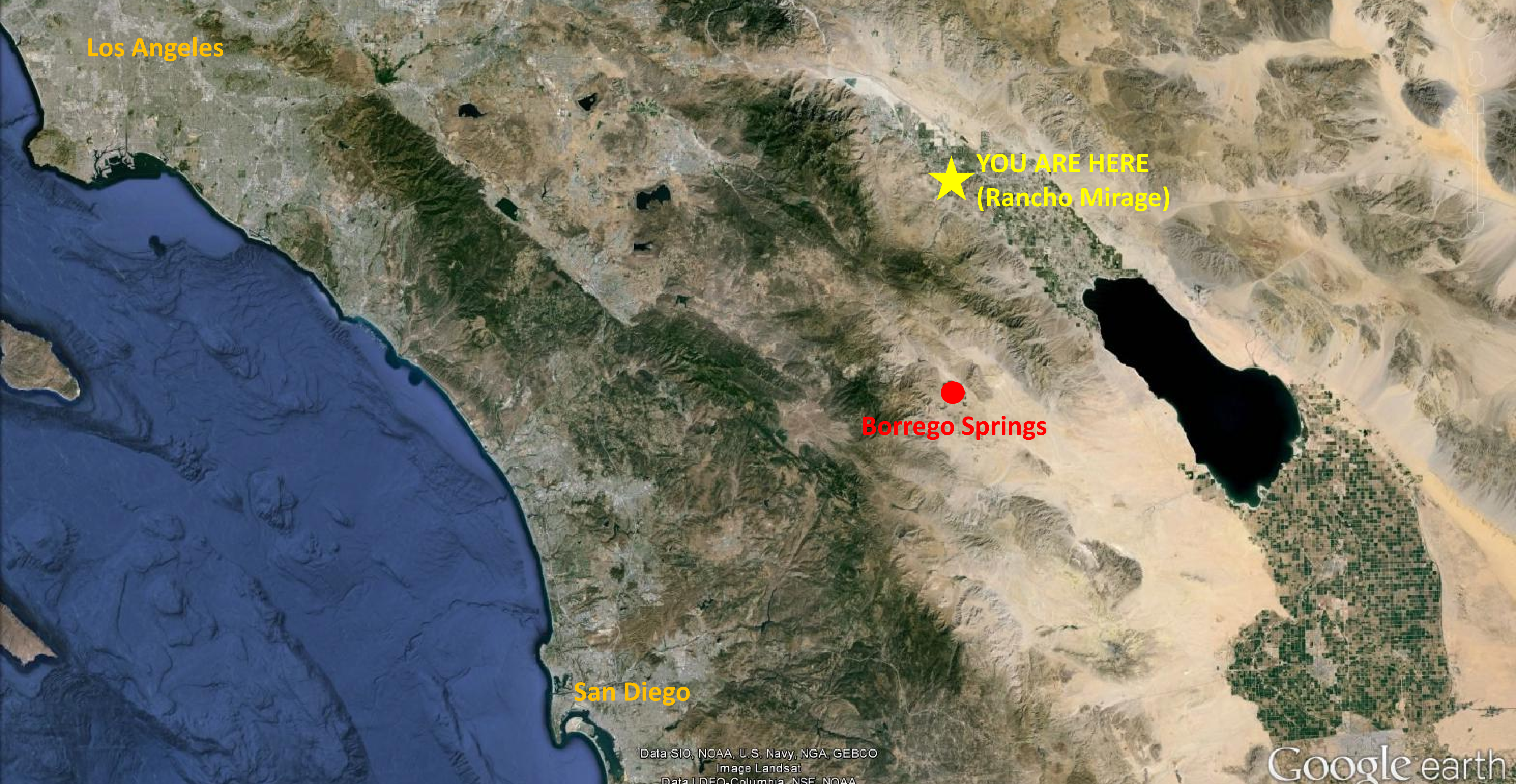
Outline

- **Where is Borrego Springs, California?**
- **History of flood events at Borrego Springs**
- **Mapping of active *vs.* inactive alluvial fan areas**
  - **Geomorphic mapping approach**
  - **Hydraulic characteristics**
  - **Integration of geomorphic- and hydraulic engineering-principles**
- **Comparison with flood hazard delineation map (FEMA FIRM)**

Where is Borrego Springs, California?

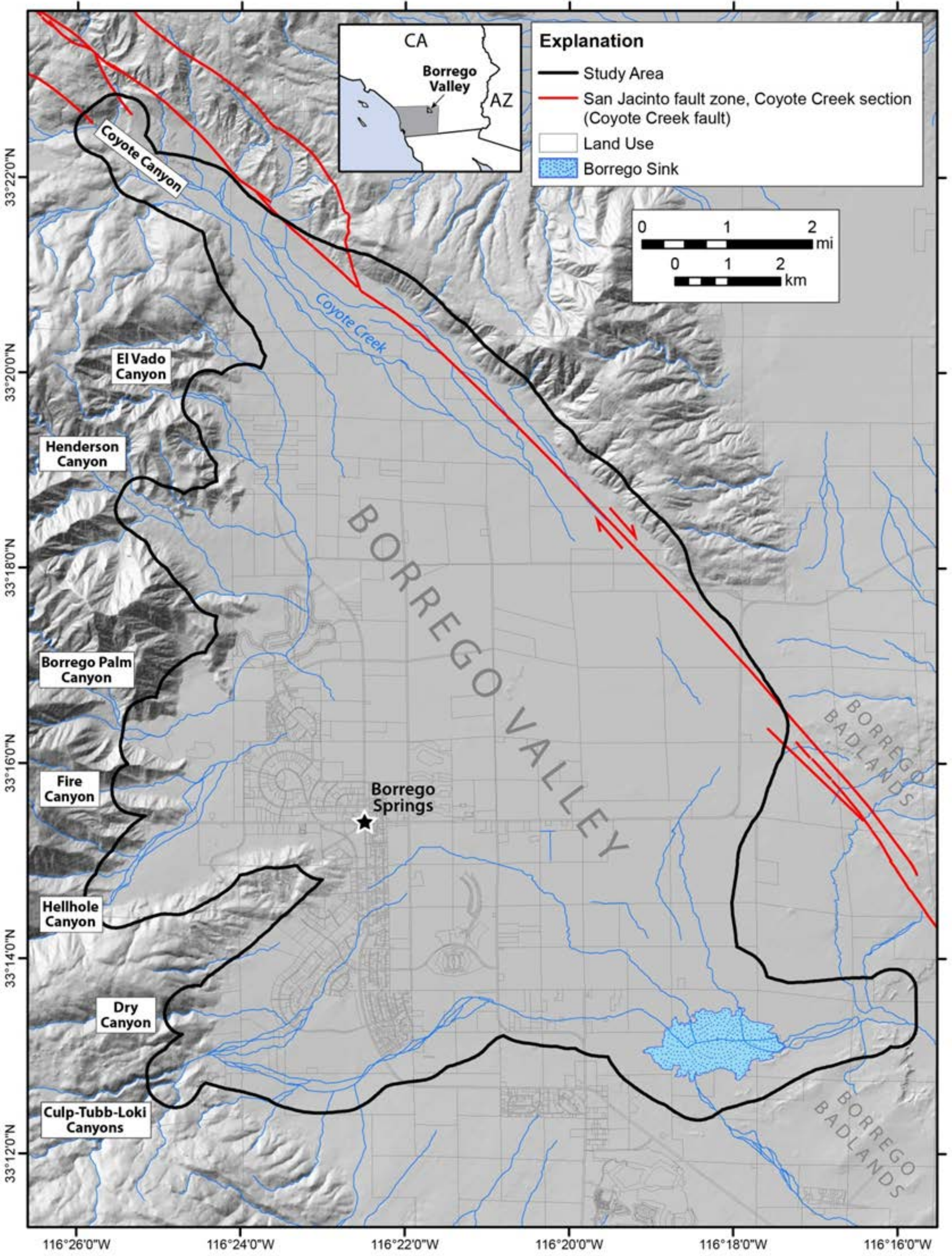


Borrego Valley

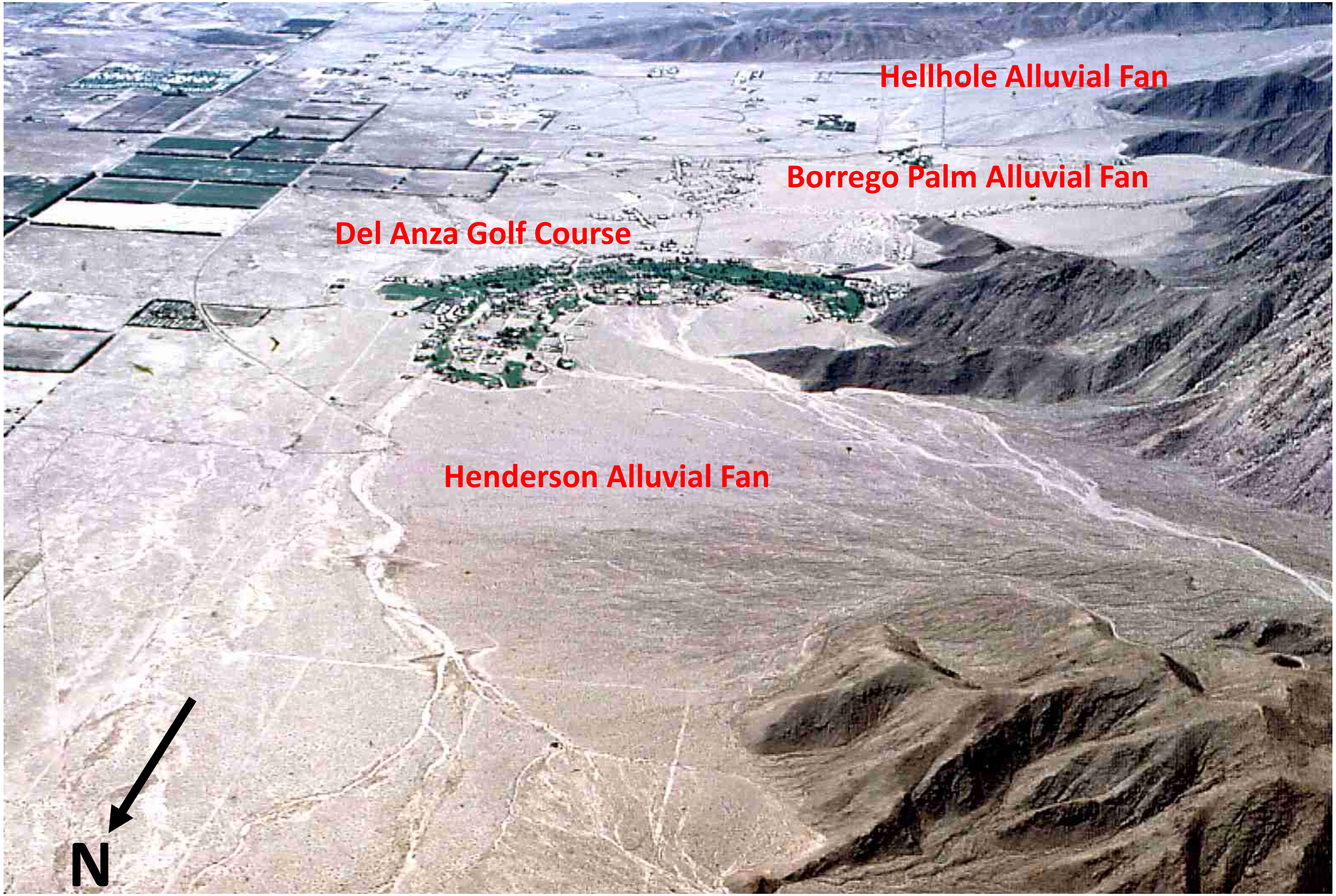
- **Alluvial Fans**
  - Coyote Canyon
  - El Vado Canyon
  - Henderson Canyon
  - Borrego Palm Canyon
  - Fire Canyon
  - Hellhole Canyon
  - Dry Canyon
  - Culp-Tubb-Loki Canyons

- **Borrego Springs**
  - Del Anza Golf Course

- **San Jacinto Fault Zone**
  - "Coyote Creek Fault"



Borrego Valley Overview



History of Recent Flooding at Borrego Springs

- 1976: Hurricane Kathleen
- 1977: Del Anzo Golf Course Flood
- 1979: Borrego Palm Canyon Flood
- 1982: Del Anzo Golf Course Flood
- 2000: Borrego Springs Flood
- 2004: Del Anzo Golf Course Flood

1976 Hurricane Kathleen Flooding

1977 Del Anza (Golf Course) Flooding

1979 Borrego Palm Canyon Flash Flood

2000 Borrego Springs Flooding

2004 Del Anza (Golf Course) Flooding

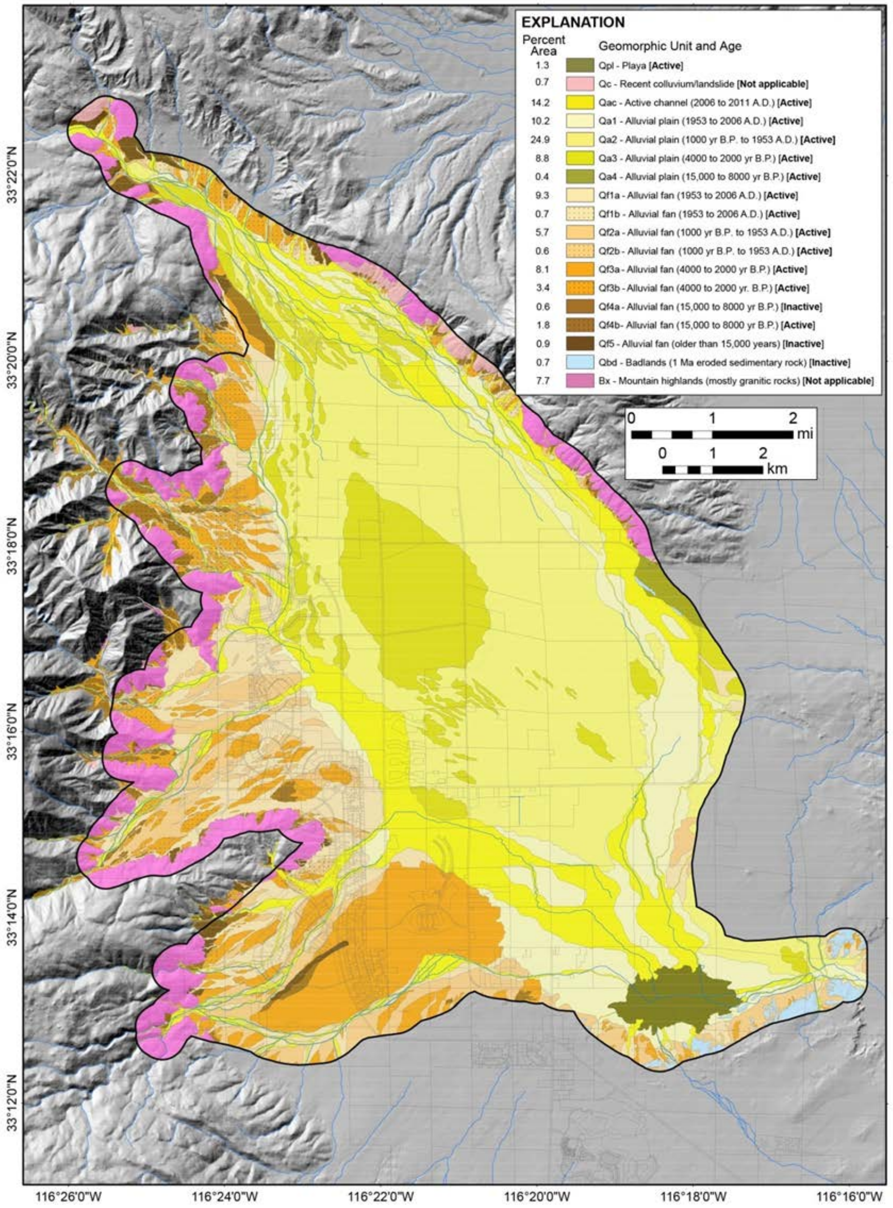
Adopted Flood Hazard Approach and Mitigation Strategies

- 1987: FEMA develops the first FIRMS for Borrego Springs
- 1989: Borrego Valley Alluvial Fan Hazard Map adopted
- 2013?: FIRMS updated
- Mitigation strategies:
  - Base Floor Elevation (BFE) above highest adjacent grade, minimum to Zone AO depth value
  - 50 % of housing lot is open space to provide drainage
  - Position structures so as not to divert damaging flows to other properties

Alluvial Fan Active *vs.* Inactive Area Mapping

- **County of San Diego requested DRI perform this work**
- **DRI instructed to follow FEMA Appendix G: "*Guidance for Alluvial Fan Flooding Analyses and Mapping*"**
- **Geomorphic mapping approach**
- **Hydraulic characteristics**
- **Integration of geomorphic- and hydraulic engineering-principles**

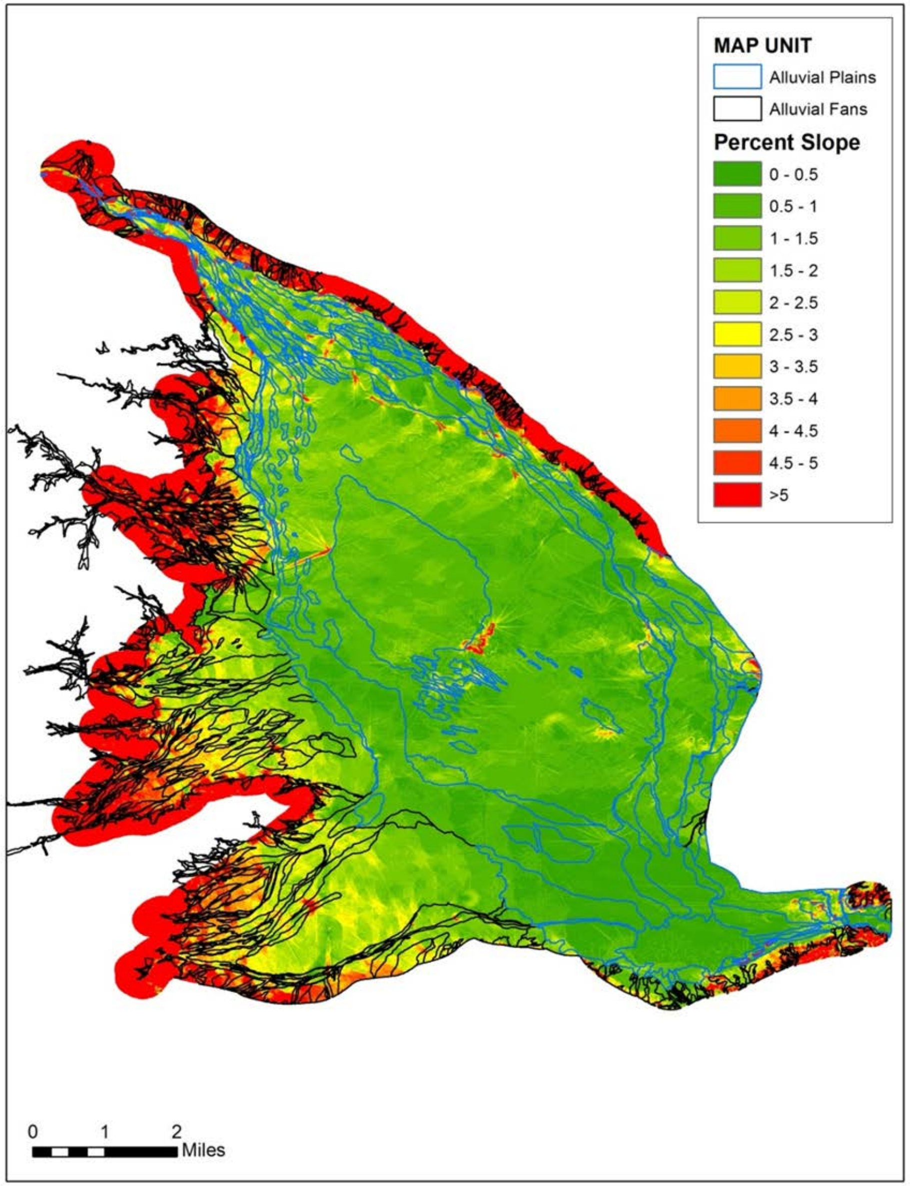
Geomorphic Map of the Borrego Valley Study Area



EXPLANATION

Percent Area	Geomorphic Unit and Age
1.3	Qpl - Playa **[Active]**
0.7	Qc - Recent colluvium/landslide **[Not applicable]**
14.2	Qac - Active channel (2006 to 2011 A.D.) **[Active]**
10.2	Qa1 - Alluvial plain (1953 to 2006 A.D.) **[Active]**
24.9	Qa2 - Alluvial plain (1000 yr B.P. to 1953 A.D.) **[Active]**
8.8	Qa3 - Alluvial plain (4000 to 2000 yr B.P.) **[Active]**
0.4	Qa4 - Alluvial plain (15,000 to 8000 yr B.P.) **[Active]**
9.3	Qf1a - Alluvial fan (1953 to 2006 A.D.) **[Active]**
0.7	Qf1b - Alluvial fan (1953 to 2006 A.D.) **[Active]**
5.7	Qf2a - Alluvial fan (1000 yr B.P. to 1953 A.D.) **[Active]**
0.6	Qf2b - Alluvial fan (1000 yr B.P. to 1953 A.D.) **[Active]**
8.1	Qf3a - Alluvial fan (4000 to 2000 yr B.P.) **[Active]**
3.4	Qf3b - Alluvial fan (4000 to 2000 yr. B.P.) **[Active]**
0.6	Qf4a - Alluvial fan (15,000 to 8000 yr B.P.) **[Inactive]**
1.8	Qf4b- Alluvial fan (15,000 to 8000 yr B.P.) **[Active]**
0.9	Qf5 - Alluvial fan (older than 15,000 years) **[Inactive]**
0.7	Qbd - Badlands (1 Ma eroded sedimentary rock) **[Inactive]**
7.7	Bx - Mountain highlands (mostly granitic rocks) **[Not applicable]**

Percent Slope Map of the Borrego Valley Study Area



Low-Gradient Alluvial Fan Landforms (Qa1 – Qa4)

Qa1

A:
- Low-gradient, alluvial plain
- Floodplain 1953-2006
- Sparse-mod vegetation
- Anastomosing and braided distributary channels

B:
- Gravelly, sand lag
- No varnish
- Stratified sediment
- No soil development

Qa4

A:
- Low-gradient, alluvial plain
- Sparse-mod vegetation with plant mounds
- Linear orientation of plant mounds controls flow direction

B:
- Gravelly, sand lag
- Incipient varnish
- Moderately-developed soil (redder soil color)

High-Gradient Alluvial Fan Landforms (Qf1a – Qf4a, Qf5)

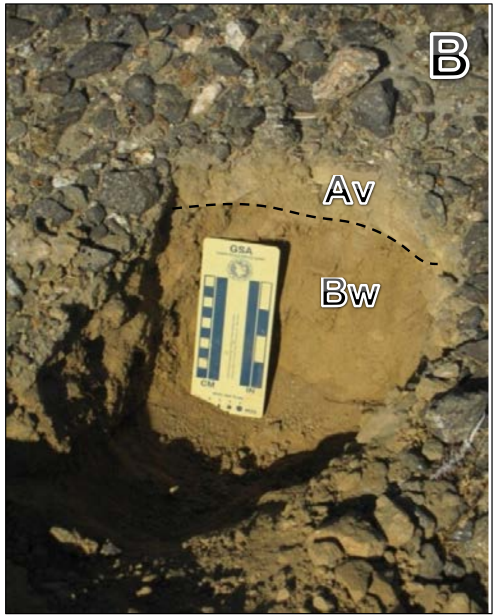
Qf2a

A:
- High-gradient, alluvial fan
- Well-vegetated
- Braided channel network

B:
- Gravelly, sand lag
- No varnish
- Non-stratified sediment
- No soil development
- Incipient soluble salt

Qf4a

A:
- High-gradient, alluvial fan
- Well-developed desert pavement and varnish
- Elevated position above channels and alluvial plain units



B:
- Desert pavement and varnish with underlying Av horizon
- Incipient varnish
- Redder soils – long-term landform stability

High-Gradient Alluvial Fan Landforms (Debris Flow: Qf1b – Qf4b)

Qf1b

A:
- High-gradient, alluvial fan
- Debris flow deposits of boulders and blocks
- Boulder levees
- Braided channel network

B:
- Imbrication of boulders
- Woody debris relict of 2005 flood
- Incipient varnish

Qf3b

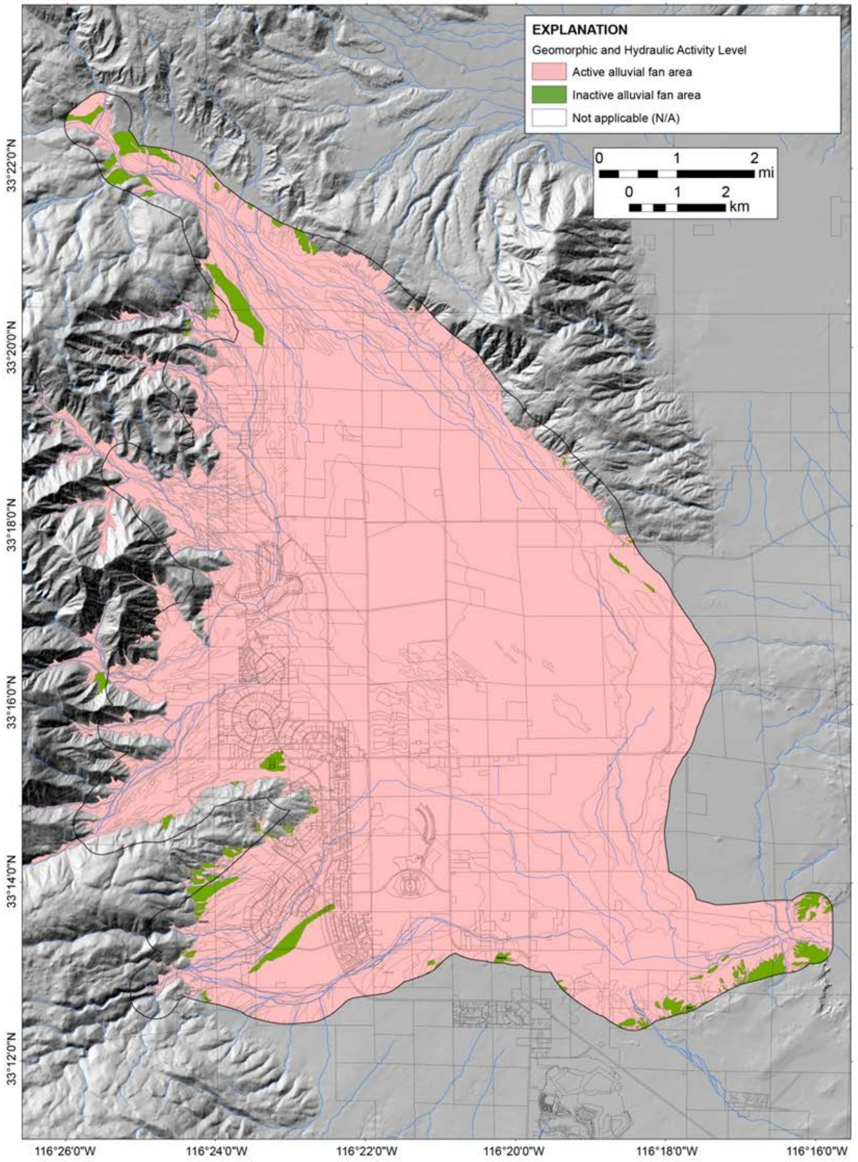
A:
- High-gradient, alluvial fan
- Debris flow deposits of boulders and blocks
- Incipient varnish

B:
- Lobes with boulder levees
- Bar-and-swale
- Braided channel network
- Incipient varnish

Hydraulic Characteristics of Active *vs.* Inactive Alluvial Fan Surfaces

Characteristic	Active Surfaces	Inactive Surfaces
Channel Continuity	Continuous; Distributary	Discontinuous; Tributary
Channel Capacity	Decreasing Downstream	No Trend
Channel Flow or Sheet Flood	Channel Flow	Sheet Flood
Debris Flow Occurrence	Yes	Minor to None
Channel Movement	Frequent	Rare

Active *versus* Inactive Alluvial Fan Areas of the Borrego Valley Study Area

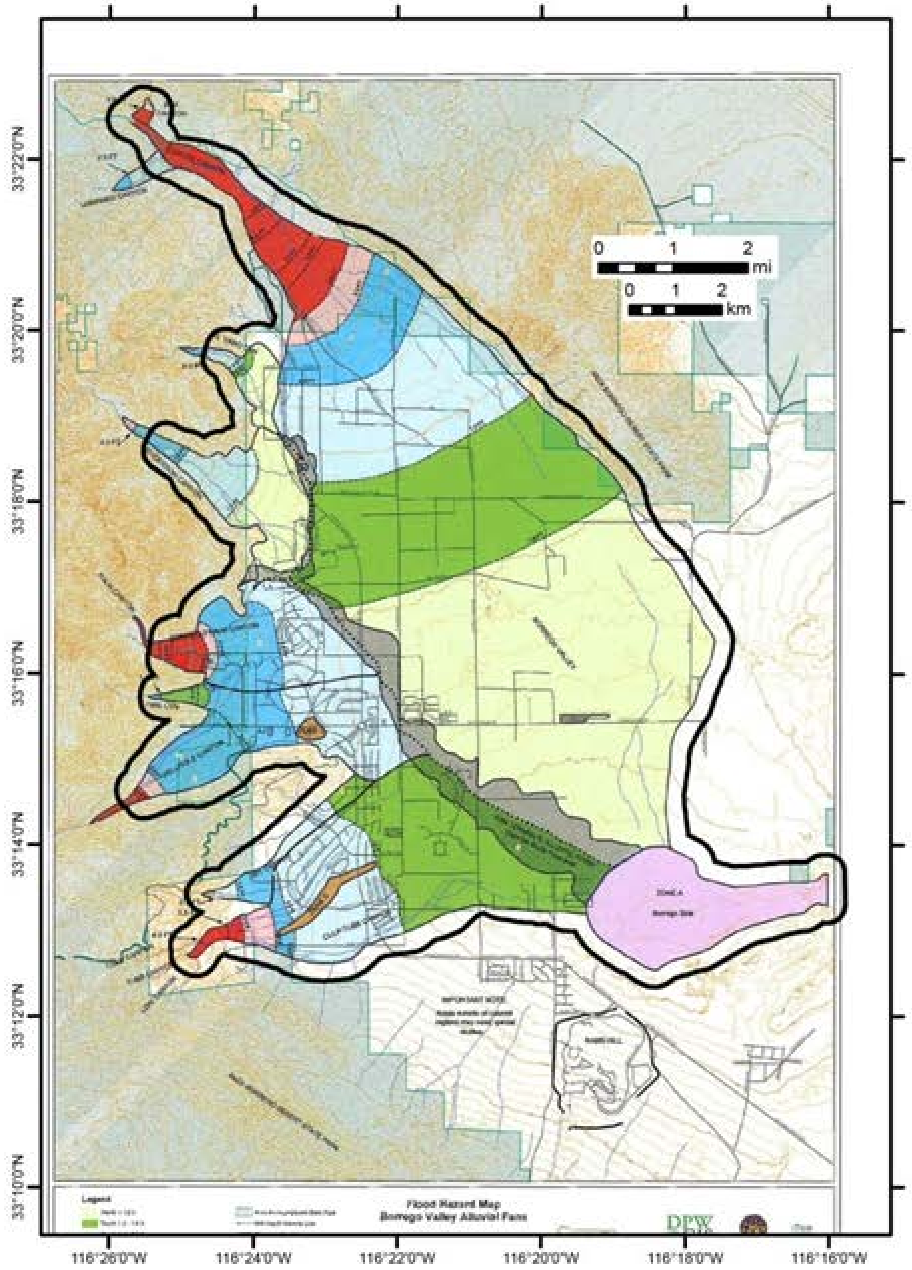


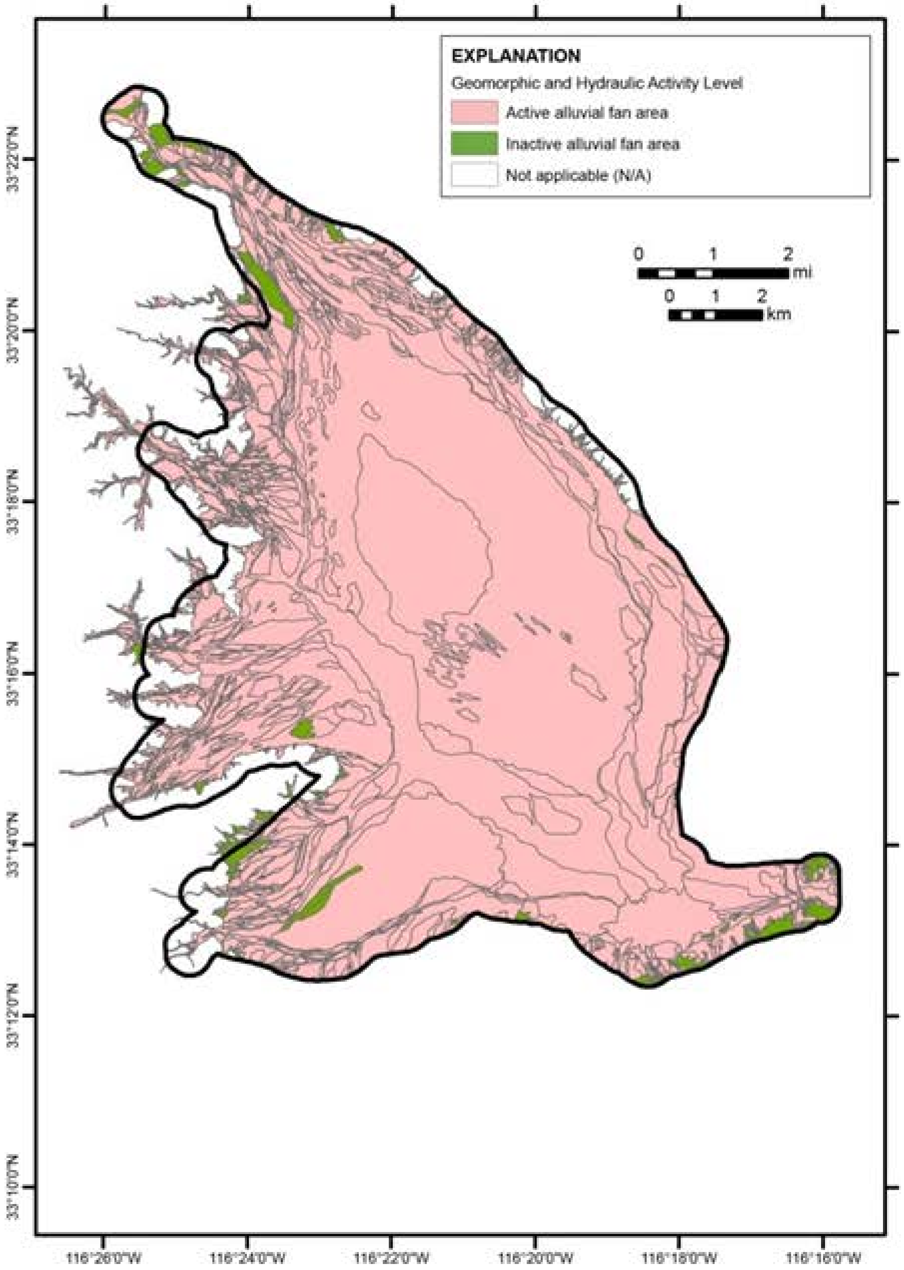
EXPLANATION

Geomorphic and Hydraulic Activity Level

- Active alluvial fan area
- Inactive alluvial fan area
- Not applicable (N/A)

Flood Hazard *versus* Active/Inactive Alluvial Fan Area Maps





Summary

- **90 % of the 61 mi^2 of the Borrego Springs Study Area contain geomorphically and hydraulically active alluvial fan landforms**

- **Results generally confirm existing flood hazard delineation map produced by Boyle Engineering (1989), which FEMA currently uses as a basis to assign flood insurance risk zones in Borrego Valley**

Adopted Mitigation Strategies

- **Mitigation strategies:**
  - Base Floor Elevation (BFE) above highest adjacent grade, minimum to Zone AO depth value
  - 50 % of housing lot is open space to provide drainage
  - Position structures so as not to divert damaging flows to other properties
  - Flood walls surrounding some structures

Adopted Mitigation Strategies

House elevated above surrounding grade

50 % open space

***note older house to the left is not elevated**

Adopted Mitigation Strategies

Flood walls surrounding houses

Adopted Mitigation Strategies

House elevated on piers

Adopted Mitigation Strategies?!

Questions?

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